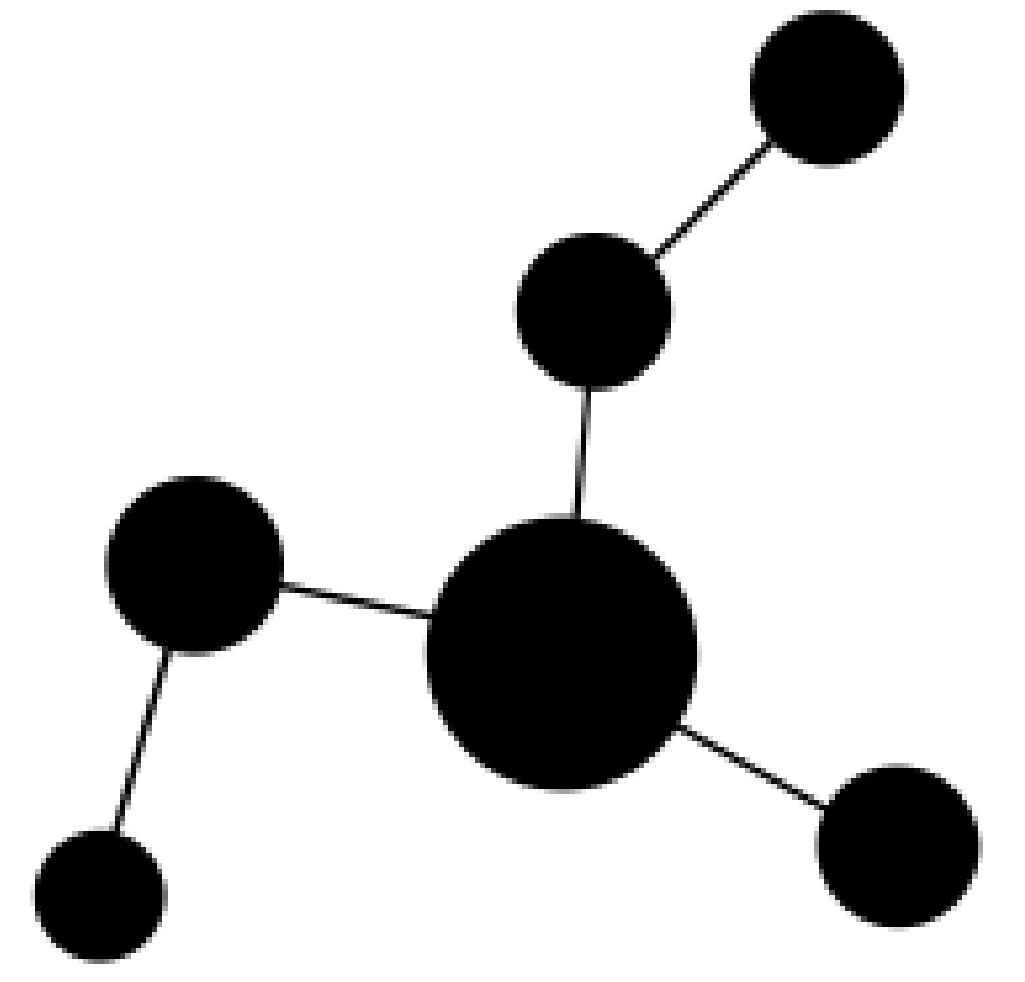


Geospatial Proximity Engine

Exact network-distance classification through safe candidate screening

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Problem

Many decisions require classifying which target locations are within a radius of one or more sources. Straight-line distance is insufficient when access depends on roads, paths, barriers, one-way restrictions or a selected travel profile.

The objective is therefore to identify source–target pairs whose shortest directed network distance is at most a radius r , and then summarize the accepted targets by count and weight. The radius may be stated in routed metres or duration; for a time radius, an assumed average movement speed supports candidate screening while OSRM evaluates the actual travel time.



Figure 1: Sources, targets and the routed network define the proximity classification problem.

Theory

Let $G = (V, E, w)$ be a directed travel graph with non-negative edge lengths. Sources are $S = (s_1, \dots, s_m)$ and targets are $T = (t_1, \dots, t_n)$. The routed distance is the directed shortest-path distance $d_G(s_i, t_j)$.

For radius $r \geq 0$, the accepted relation is

$$R_r = \{(i, j) : d_G(s_i, t_j) \leq r\}.$$

Let $\delta(s, t)$ denote great-circle distance. For every feasible path $P = (v_0 = s, \dots, v_k = t)$, the triangle inequality gives

$$\delta(s, t) \leq \sum_{\ell=1}^k \delta(v_{\ell-1}, v_\ell) \leq \sum_{\ell=1}^k w(v_{\ell-1}, v_\ell).$$

Minimizing over all feasible paths yields

$$\delta(s, t) \leq d_G(s, t).$$

Consequently, $\delta(s, t) > r$ implies $d_G(s, t) > r$. Such a pair cannot be accepted and may be removed before routing. Candidate screening is therefore exact: it reduces OSRM work without removing any true network-distance match.



Figure 2: Road-following relations from two sources to accepted and rejected candidates.

Four-Step Algorithm

- Candidate selection.** Retain (i, j) only when the geographic lower bound satisfies $\delta(s_i, t_j) \leq r$.
- Directed distance calculation.** Use OSRM to evaluate $D_{ij} = d_G(s_i, t_j)$ for candidate pairs in the conceptual $m \times n$ source-to-target matrix.
- Radius decision.** Accept candidate (i, j) when $D_{ij} \leq r$; otherwise classify it as outside the routed radius.
- Aggregation.** Return accepted-relation count, summed target weight, unique-target count and unique summed weight.

The distinction between relation-based and unique measures is important. A target reached from multiple sources contributes once per relation to the first set of measures, but only once to unique geographic coverage.

Implementation

The implementation separates mathematical screening from network routing. Python controls the workflow, while OSRM evaluates directed shortest paths for the selected travel profile.

Component	Responsibility	Output
Preprocessing	Validate coordinates and weights; compute the Haversine lower bound	Candidate index pairs
Distance engine	Query OSRM Table in bounded chunks while preserving direction	Source-to-target distance blocks
Classification	Apply the radius rule and aggregate accepted relations	Matrix, counts and weighted summaries
Visualisation	Request OSRM route geometry and render it with Folium	Interactive route map

Table 1: Implementation responsibilities and data products.

The computational interface returns a NumPy distance matrix, explicit accepted relations and both relation-based and unique weighted summaries. The same code supports walking, driving and cycling by changing only the OSRM profile, with the radius expressed either as distance or duration.

Applications

The engine is a reusable classification component. Business meaning is supplied through the choice of sources, targets, travel profile, radius and target weight.

Use case	Sources	Decision output
Address coverage	Candidate sites	Reachable addresses and weighted potential
Competitive context	Own or proposed stores	Comparable stores within n routed metres
Lead classification	Lead locations	Nearby targets, unique coverage and total weight
Service planning	Depots or facilities	Network-accessible demand by travel profile

Table 2: Representative applications of the same source–target relation.

Contribution

- Exact candidate pruning justified by the triangle inequality.
- Directed route-distance classification rather than circular buffering.
- Scalable OSRM table queries restricted to viable candidates.
- Relation-based and unique weighted summaries for decision support.
- A reusable implementation across sites, stores, addresses and leads.

Key Message

Geographic distance provides a safe lower bound; network distance makes the decision. The combination preserves exactness while avoiding unnecessary routing queries.